

NAME	KEN KIBUGI MUIRURI
TRACKING NUMBER	CS-SA12-26014
COHORT	CYBERSHUJAA COHORT 1-2026
TASK	INTRO TO LOG ANALYSIS REPORT

Contents

1.0	Log Analysis Basics.....	2
2.0	Investigation Theory	2
3.0	Detection Engineering	3
4.0	Automated vs. Manual Analysis	3
5.0	Log Analysis Tools: Command Line.....	3
6.0	Log Analysis Tools: Regular Expressions.....	6
7.0	Log Analysis Tools: CyberChef	6
8.0	Log Analysis Tools: Yara and Sigma	9
	Conclusion	9

INTRO TO LOG ANALYSIS REPORT

Introduction

In this report, I explored the fundamentals of log analysis and how to work with different tools and techniques to extract, parse, and interpret log data. I worked with command-line filtering, regular expressions, CyberChef recipes to see how logs can reveal meaningful insights and potential security issues.

1.0 Log Analysis Basics

I learned that logs are time-based records of events in systems and networks, and analyzing them helps understand what has happened. I also understood that different types of logs provide valuable insights for troubleshooting, detecting threats, and maintaining security.

2.0 Investigation Theory

I learned that effective log investigation involves creating timelines to understand the sequence of events and correlate data from multiple sources. I also understood the importance of visualization, alerting, and using threat intelligence to quickly identify and respond to security incidents.

Question1: What's the term for a consolidated chronological view of logged events from diverse sources, often used in log analysis and digital forensics?

Answer: **Super Timeline**

Question2: Which threat intelligence indicator would 5b31f93c09ad1d065c0491b764d04933 and 763f8bdbc98d105a8e82f36157e98bbe be classified as?

Answer: **File Hashes**

3.0 Detection Engineering

I learned that effective detection in log analysis involves knowing common log locations, spotting abnormal user behavior, and identifying attack signatures such as SQL injection, XSS, and path traversal attempts. Recognizing these patterns helps security teams respond quickly to threats.

Question1: What is the default file path to view logs regarding HTTP requests on an Nginx server?

Answer: **`/var/log/nginx/access.log`**

Question2: A log entry containing %2E%2E%2F%2E%2E%2Fproc%2Fself%2Fenviron was identified. What kind of attack might this infer?

Answer: **Path Traversal**

4.0 Automated vs. Manual Analysis

I learned that automated log analysis uses tools and AI to quickly detect patterns, while manual analysis allows for deeper, context-rich investigations. Both methods are important for effective threat detection.

Question1: A log file is processed by a tool which returns an output. What form of analysis is this?

Answer: **Automated**

Question2: An analyst opens a log file and searches for events. What form of analysis is this?

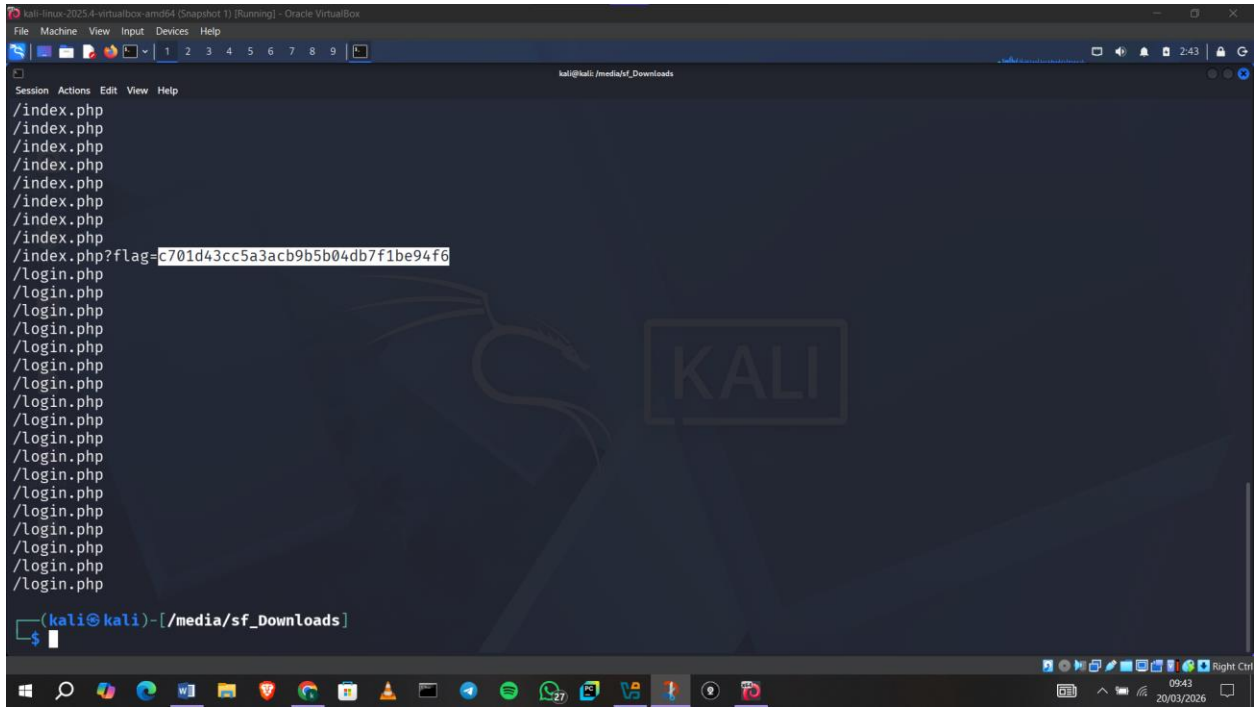
Answer: **Manual**

5.0 Log Analysis Tools: Command Line

I learned that command-line tools like grep, awk, and cut can be used to quickly filter, search, and analyze log data. These tools allow efficient investigation and pattern detection even without advanced log analysis platforms.

Question1: Use cut on the apache.log file to return only the URLs. What is the flag that is returned in one of the unique entries?

Answer: I ran `cut -d ' ' -f 7 apache-1691435735822.log | sort -n` and found the flag:
c701d43cc5a3acb9b5b04db7f1be94f6



```
kali-linux-2019.4-vmtoolsd-undf64 (Snapshot 1) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4 5 6 7 8 9
kali@kali: /media/sf_Downloads
Session Actions Edit View Help
/index.php
/index.php
/index.php
/index.php
/index.php
/index.php
/index.php
/index.php
/index.php?flag=c701d43cc5a3acb9b5b04db7f1be94f6
/login.php
/login.php
/login.php
/login.php
/login.php
/login.php
/login.php
/login.php
/login.php
/login.php
/login.php
/login.php
/login.php
/login.php
/login.php
/login.php
/login.php
/login.php
/login.php
/login.php
/login.php
(kali@kali) - [/media/sf_Downloads]
$
```

Question2: In the apache.log file, how many total HTTP 200 responses were logged?

Answer: I ran `awk '$9 == 200' apache-1691435735822.log | wc -l` and found 52

```
kali-linux-2023.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali:/media/sf_downloads
Session Actions Edit View Help
145.76.33.201 - - [31/Jul/2023:12:34:34 +0000] "GET /contact.php HTTP/1.1" 200 5678 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.330 Safari/537.36"
110.122.65.76 - - [31/Jul/2023:12:34:33 +0000] "GET /about.php HTTP/1.1" 200 9876 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.100 Safari/537.36"
176.145.201.99 - - [31/Jul/2023:12:34:31 +0000] "GET /index.php?flag=c701d43cc5a3acb9b5b04db7f1be94f6 HTTP/1.1" 200 1234 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.270 Safari/537.36"
104.76.29.88 - - [31/Jul/2023:12:34:30 +0000] "GET /contact.php HTTP/1.1" 200 5678 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.350 Safari/537.36"
76.89.54.221 - - [31/Jul/2023:12:34:28 +0000] "GET /login.php HTTP/1.1" 200 7890 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.330 Safari/537.36"
145.76.33.201 - - [31/Jul/2023:12:34:27 +0000] "GET /index.php HTTP/1.1" 200 4321 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.330 Safari/537.36"
110.122.65.76 - - [31/Jul/2023:12:34:26 +0000] "GET /contact.php HTTP/1.1" 200 7890 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.350 Safari/537.36"
176.145.201.99 - - [31/Jul/2023:12:34:24 +0000] "GET /login.php HTTP/1.1" 200 1234 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.90 Safari/537.36"
104.76.29.88 - - [31/Jul/2023:12:34:23 +0000] "GET /index.php HTTP/1.1" 200 5678 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.81 Safari/537.36"
76.89.54.221 - - [31/Jul/2023:12:34:21 +0000] "GET /about.php HTTP/1.1" 200 1234 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.85 Safari/537.36"
145.76.33.201 - - [31/Jul/2023:12:34:20 +0000] "GET /login.php HTTP/1.1" 200 5678 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/94.0.4606.90 Safari/537.36"
awk: cmd. line:1: fatal: cannot open file `c' for reading: No such file or directory

(kali@kali)~/media/sf_downloads
$ awk 'NR == 200' apache-1691435735822.log | wc -l
52

(kali@kali)~/media/sf_downloads
$
```

Question3: In the apache.log file, which IP address generated the most traffic?

Answer: I ran `cut -d ' ' -f 1 apache-1691435735822.log | sort -n -r | uniq -c` and found the IP as **145.76.33.201**

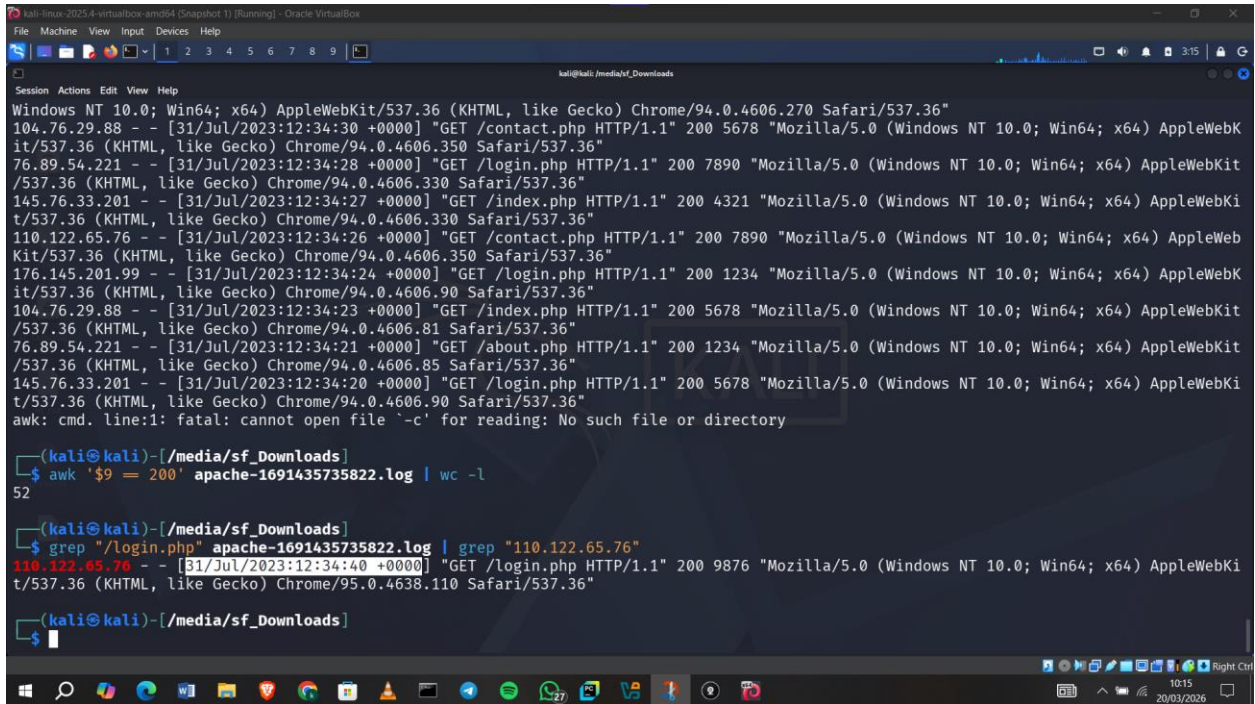
```
kali-linux-2023.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali:/media/sf_downloads
/login.php
/login.php
/login.php

(kali@kali)~/media/sf_downloads
$ cut -d ' ' -f 1 apache-1691435735822.log | sort -n -r | uniq -c
 1 221.90.64.76
 1 211.87.186.35
 1 203.78.122.88
 6 203.64.78.90
 1 203.0.113.42
 1 202.176.73.99
 1 201.39.104.77
 1 200.89.134.22
 1 192.168.45.99
 1 189.76.230.44
 1 188.99.145.201
 1 188.96.105.221
 1 188.145.201.99
 1 185.76.230.45
 2 178.53.64.100
 1 177.34.78.90
 6 176.145.201.99
 1 156.98.34.12
 1 153.47.106.221
 8 145.76.33.201
 1 141.89.23.76
 6 128.45.76.66
 1 122.65.187.55
 1 120.54.86.23
```

Question4: What is the complete timestamp of the entry where 110.122.65.76 accessed /login.php?

Answer: I ran **grep "/login.php" apache-1691435735822.log | grep "110.122.65.76"** and found the timestamp as: **31/Jul/2023:12:34:40 +0000**



```
kali@kali:~/media/sf_downloads$ cat apache-1691435735822.log | wc -l
52
(kali@kali)~/media/sf_downloads$ awk 'NR == 200' apache-1691435735822.log | wc -l
1
(kali@kali)~/media/sf_downloads$ grep "/login.php" apache-1691435735822.log | grep "110.122.65.76"
110.122.65.76 - - [31/Jul/2023:12:34:40 +0000] "GET /login.php HTTP/1.1" 200 9876 "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/95.0.4638.110 Safari/537.36"
```

6.0 Log Analysis Tools: Regular Expressions

I learned that regular expressions are used to define patterns for searching and extracting specific data from log files. They help filter, parse, and structure logs, making analysis more efficient and accurate.

Question1: How would you modify the original grep pattern above to match blog posts with an ID between 20-29?

Answer: **post=2[0-9]**

Question2: What is the name of the filter plugin used in Logstash to parse unstructured log data?

Answer: **Grok**

7.0 Log Analysis Tools: CyberChef

I learned that CyberChef is a versatile tool used to analyze, decode, and process data using customizable operations. It can also apply regex to extract specific information from log files, making analysis more efficient.

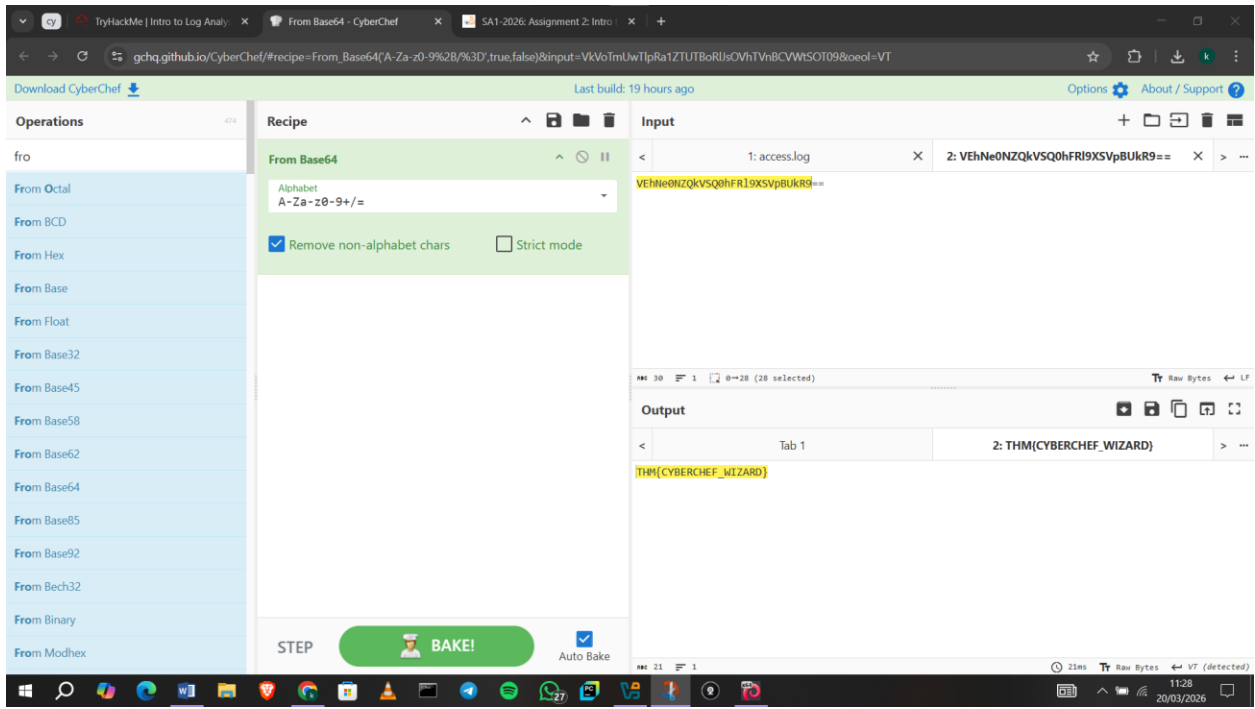
Question1: Upload the log file named "access.log" to CyberChef. Use regex to list all of the IP addresses. What is the full IP address beginning in 212?

Answer: I used `\b([0-9]{1,3}\.){3}[0-9]{1,3}\b` and found it: **212.14.17.145**

The screenshot shows the CyberChef web interface. On the left, the 'Operations' sidebar is visible with 'Regular expression' selected. The 'Recipe' panel shows a 'Regular expression' recipe with the user-defined regex `\b([0-9]{1,3}\.){3}[0-9]{1,3}\b`. The 'Input' panel displays the contents of 'access.log', which includes several log entries with IP addresses. The 'Output' panel shows the result of the regex operation, listing all IP addresses found in the log, with '212.14.17.145' at the top. The 'File details' panel on the right shows the file name 'access.log', size '19,051 bytes', and type 'unknown'.

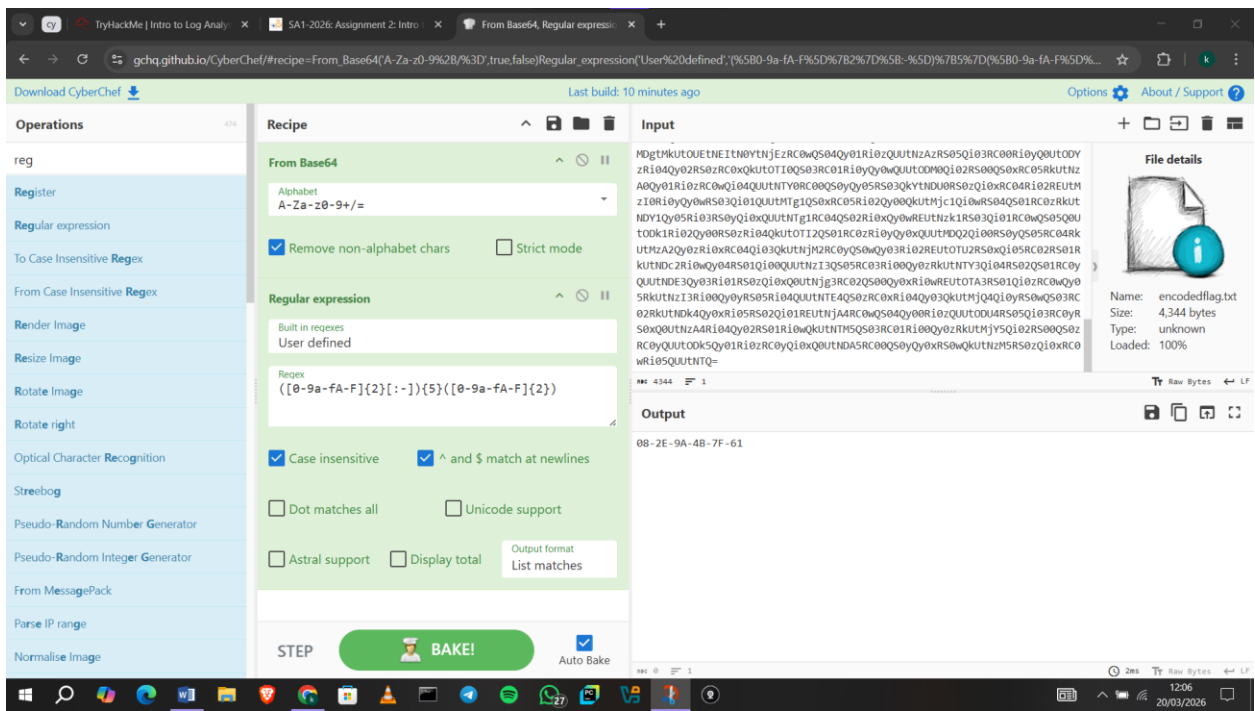
Question2: Using the same log file from Question #2, a request was made that is encoded in base64. What is the decoded value?

Answer: **THM{CYBERCHEF_WIZARD}**



Question3: Using CyberChef, decode the file named "encodedflag.txt" and use regex to extract by MAC address. What is the extracted value?

Answer: I used the regular expression `(([0-9a-fA-F]{2})[:-]){5}([0-9a-fA-F]{2})` and found the MAC address: **08-2E-9A-4B-7F-61**



8.0 Log Analysis Tools: Yara and Sigma

I learnt that Both Sigma and YARA provide powerful mechanisms to automate log monitoring and analysis. Sigma is more suited for structured SIEM environments, while YARA offers flexibility for scanning raw logs or files. Together, they enhance an analyst's ability to detect anomalies and potential security threats efficiently.

Question1: What languages does Sigma use?

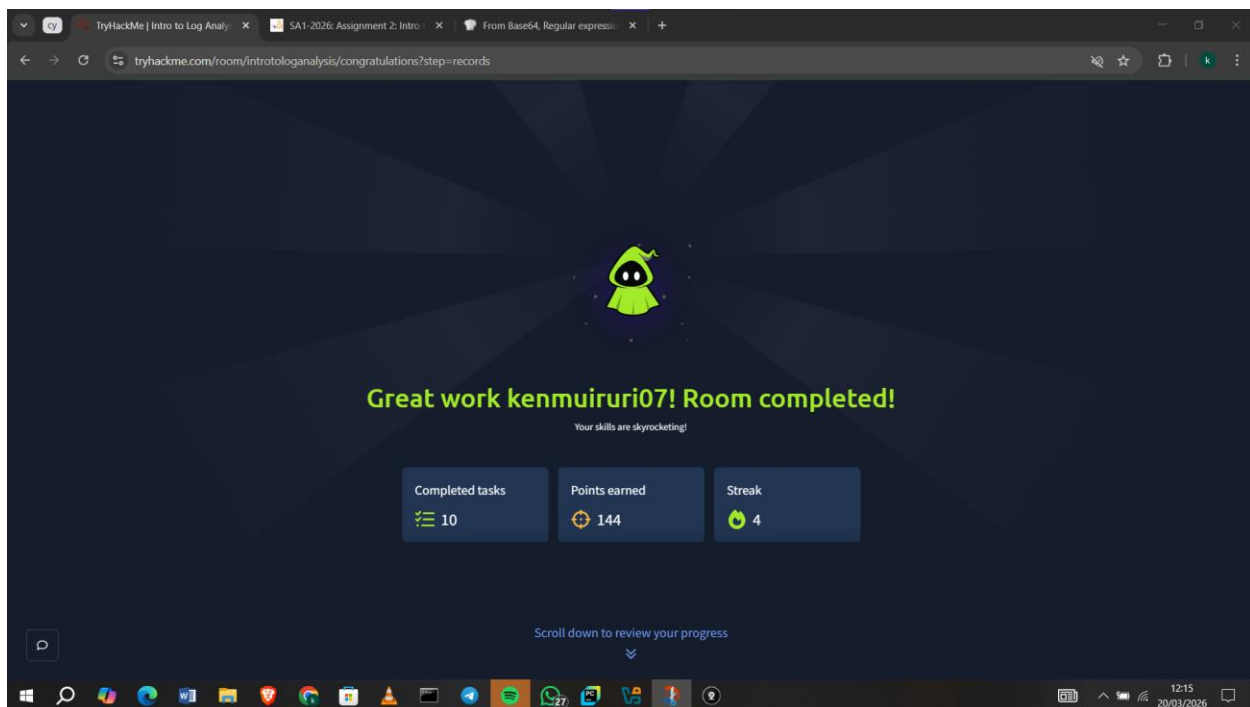
Answer: **YAML**

Question2: What keyword is used to denote the "title" of a Sigma rule?

Answer: **title**

Question3: What keyword is used to denote the "name" of a rule in YARA?

Answer: **rule**



Conclusion

Through this task, I learned how powerful log analysis can be in detecting events, identifying patterns, and automating the monitoring of systems. Using tools like grep, CyberChef, YARA, and Sigma, I was able to efficiently analyze logs and better understand how to uncover anomalies and strengthen security monitoring.